

Lecture 9: Statistics

CMSC/DATA 320: Introduction to Data Science

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Announcements

- PA2, WA2 due yesterday
 - Quiz 4 grades and solutions will be out today
 - PA3, WA3 out tomorrow
- 

Today we'll discuss

- Some statistics and probability topics



Descriptive Statistics / Summary Statistics

Data can be presented in many different formats, but, **what are the main characteristics that describe the data set?**

Descriptive statistics involve summarizing, organizing, and presenting data (the features of a dataset) meaningfully and concisely.

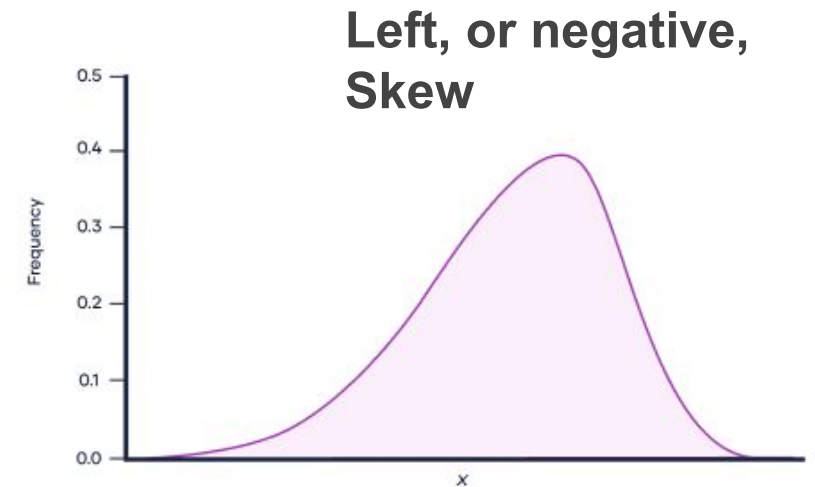
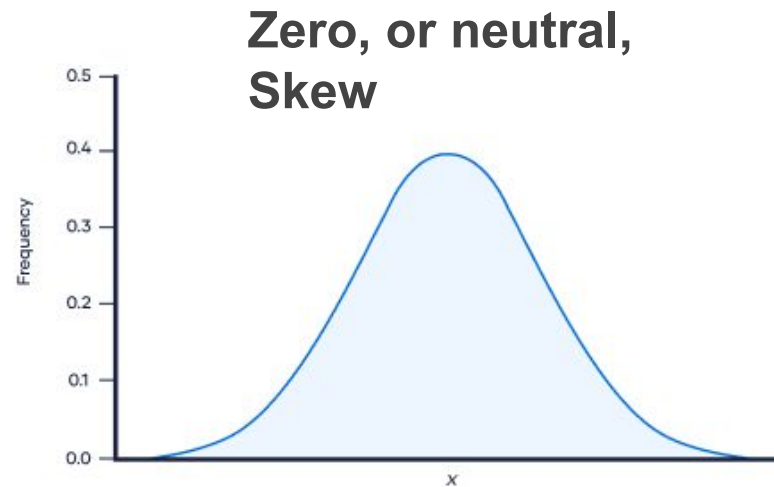
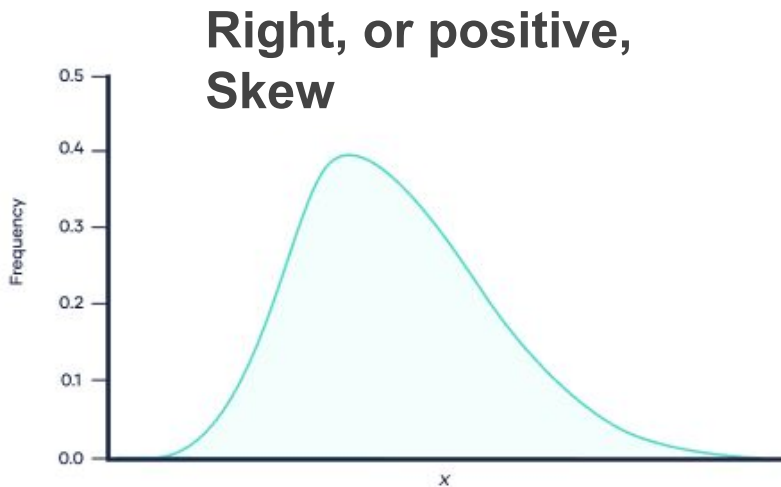
- **Focus:** Describes and analyzes dataset's main features *without* generalizing to larger populations.
- **Enables insights:** Helps researchers understand patterns, trends, and distributions *within* the dataset

What *are* the mean, standard deviation, and skew?



Skew

- Skew, or skewness, is a measure of **asymmetry** for a probability distribution
 - Deals with the distribution of random variable about its mean
- A normal distribution has a skew of zero
-  Note! Common usage of “skew” is the **opposite** of statistical usage:



Measures of Central Tendency

Central tendency represents the center, or middle, of a set of values

We use 3 common measures:

Mode

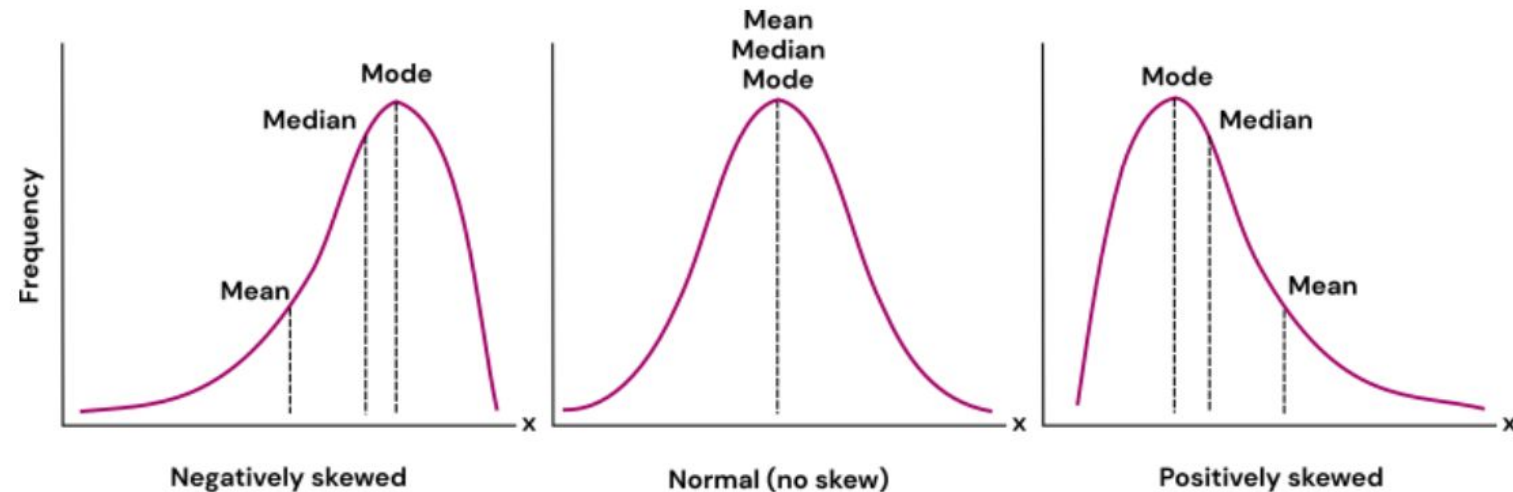
the **most frequent** value

Median

the **middle** number in an ordered dataset

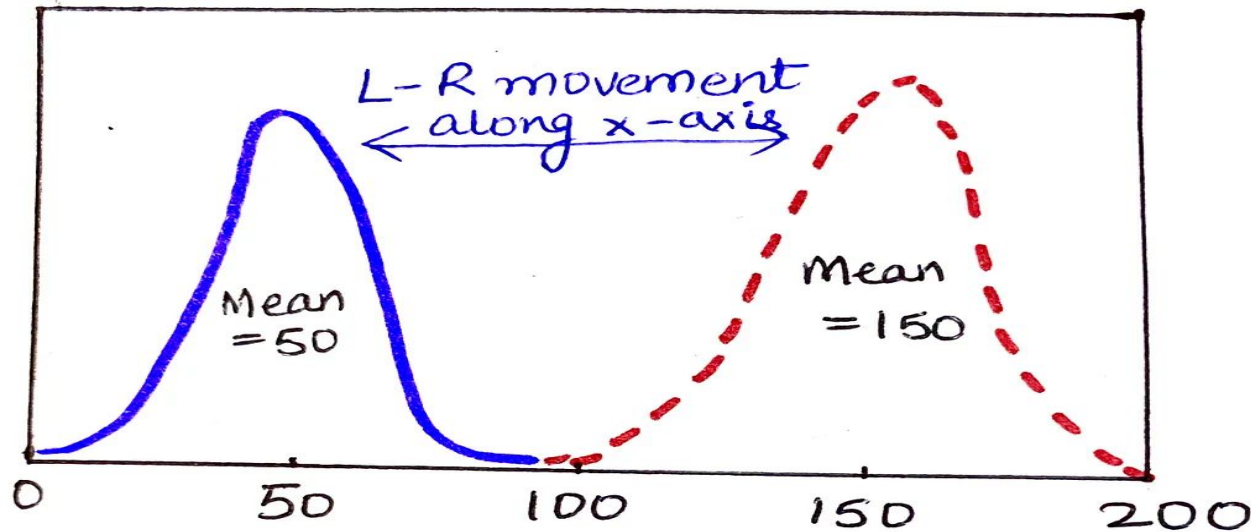
Mean

the **average** of the values
(sum of all values divided by the total number of values)



More on Mean (μ)

- Average of all values.
- In a normal distribution:
 - μ is the central tendency
 - mean = median = mode
 - This is why we only need μ and σ
- Change in mean results in:
 - horizontal shifting of the curve along the x-axis



$$\mu = \frac{\sum_{i=1}^n x_i}{n}$$

Measures of Dispersion

Help us understand the variability of the data

- How smushed or spread out the random variable distribution is

Range	Max - Min	What interval the data lives on along the the x axis
Variance	$V = \sigma^2 = \frac{\sum (X - \mu)^2}{N}$	Average degree to which each value differs from the mean
Standard Deviation	$\sigma = \sqrt{V} = \sqrt{\frac{\sum (X - \mu)^2}{N}}$	Spread of values from the mean

Note: these are *population* formulas (will discuss in a moment)

Formulas: Measures of Variance/Spread

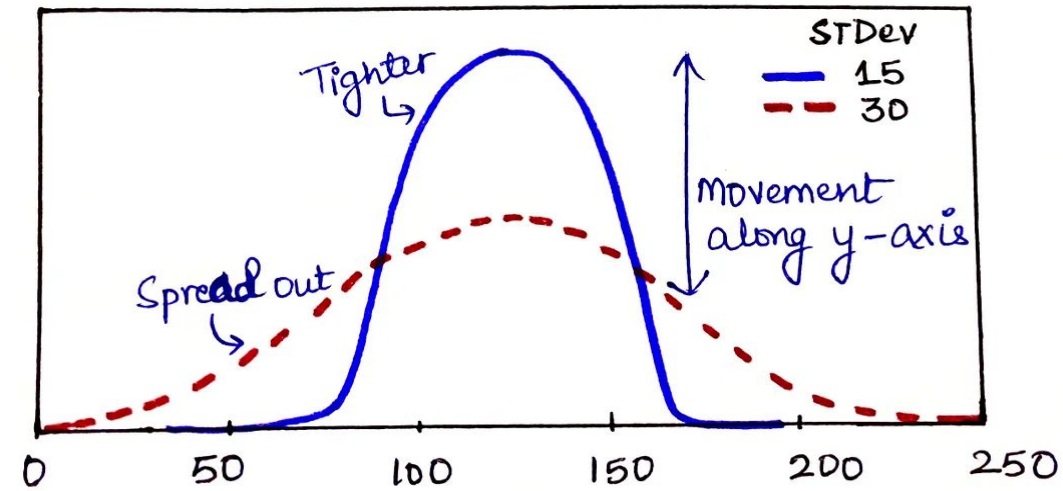
Calculating standard deviation depends on whether the values are for the **entire population** or are from **a sample**.

The calculations are similar, but *not identical*.

	Sample	Population
Variance	$\hat{V} = s^2 = \frac{\sum_{i=1}^n (X_i - \bar{x})^2}{n - 1}$	$V = \sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$
Standard Deviation	$s = \sqrt{\hat{V}} = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{x})^2}{n - 1}}$	$\sigma = \sqrt{V} = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}}$

Standard Deviation

- More convenient to use σ than V
 - Easier to see and apply
 - Same units as the random variable
 - Variance involves squaring differences, making comparisons across datasets with different units or scales challenging.
 - Can talk about “how many” σ ’s there are
- For normal distributions:
 - Tells us how far values are from the mean
 - Determines the width of the curve
 - Tighter the curve (lesser the width) → taller the height
 - More spread-out the curve (higher the width) → shorter the height
- Widely used

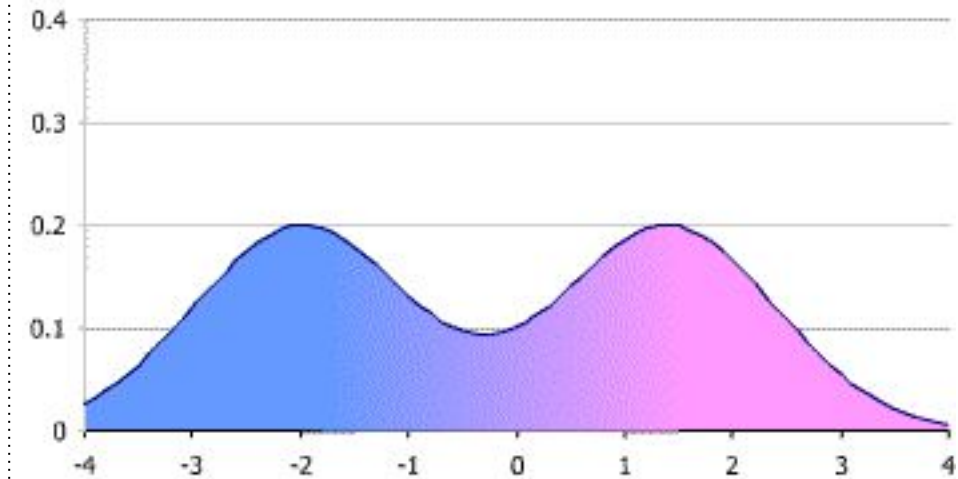


Modality

Modality: Refers to the number of peaks or modes in a distribution. A distribution can be:

- **Unimodal** (one peak)
- **Bimodal** (two peaks)
- **Trimodal** (three peaks)
- **Multimodal** (more than three peaks)

depending on how the modes are distributed within the data.



Useful because it provides insights into the underlying structure (pattern or cluster) and characteristics of the data

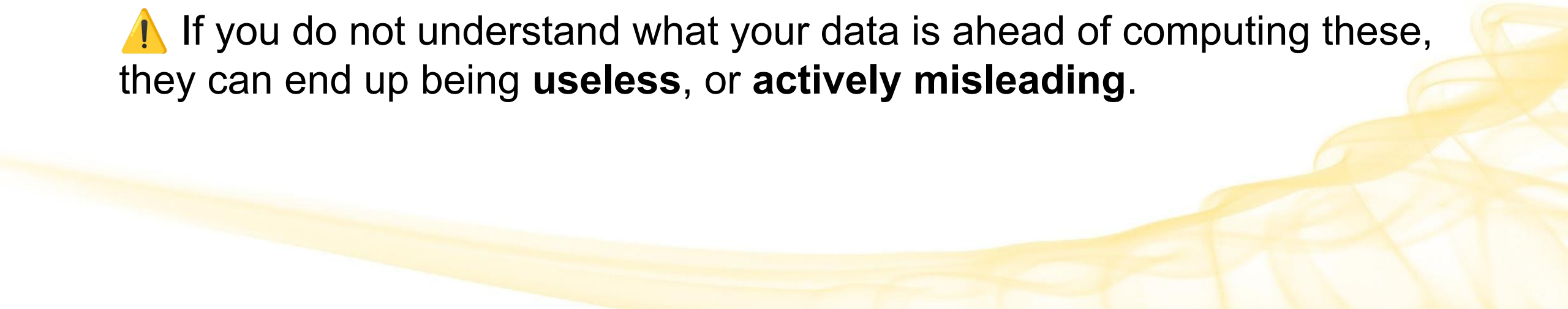
A word of caution

Descriptive / summary statistics are simple but **powerful!**

- Can tell us a lot about our data set
- Can help us communicate in a known/common way
- Very useful for quickly understanding what's going on

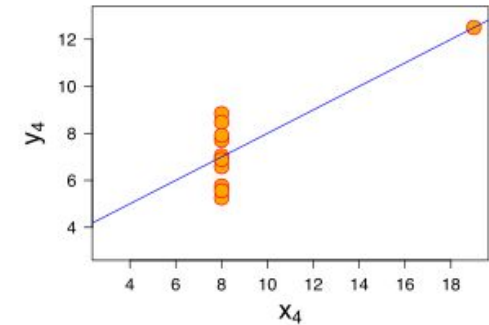
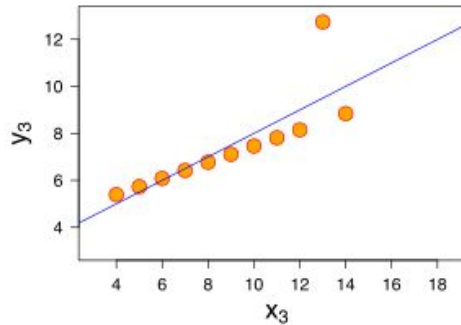
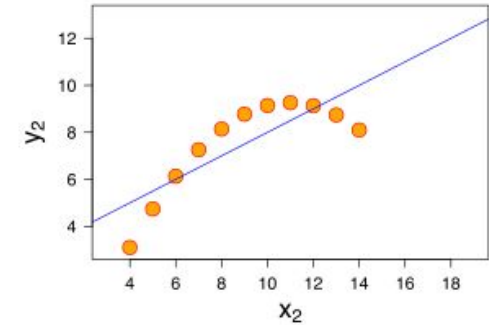
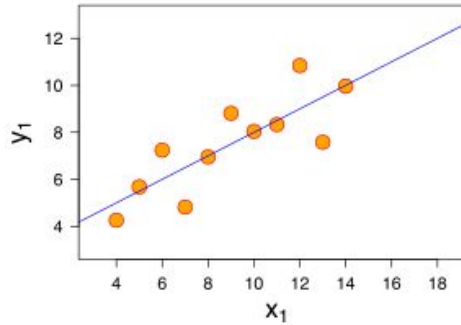
However, they are **not a substitute** for critical thinking! They are merely an aid.

⚠ If you do not understand what your data is ahead of computing these, they can end up being **useless**, or **actively misleading**.

A decorative graphic consisting of several overlapping, flowing yellow lines that swirl and curve across the bottom right portion of the slide, creating a sense of movement and depth.

Why not to rely solely on summary statistics

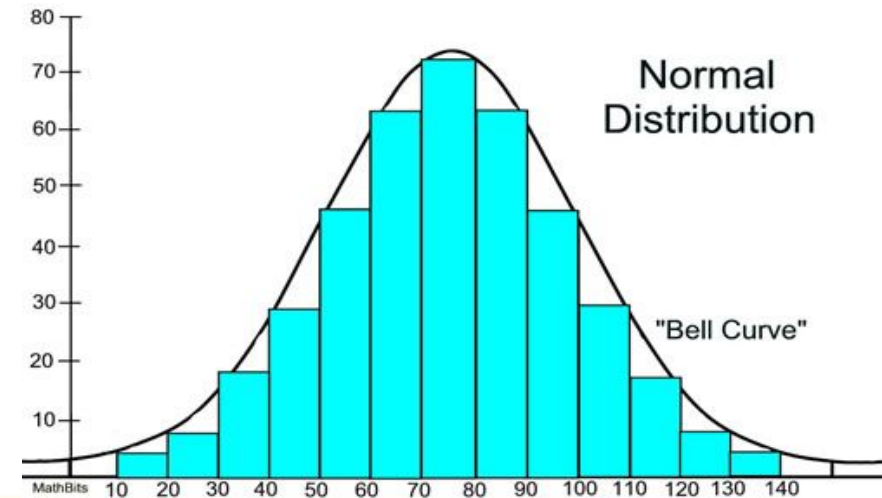
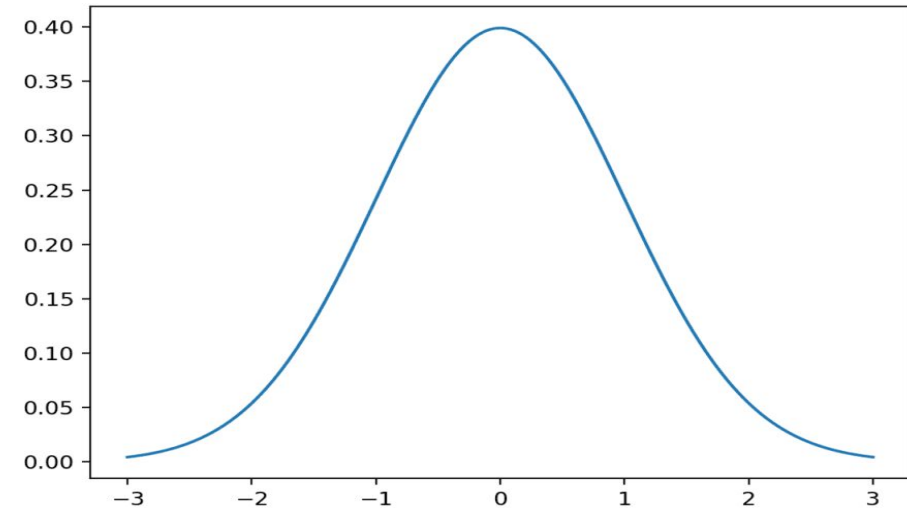
All four of these have the **same mean and variance**, but are clearly generated by four different processes.



We also need to understand the overall distribution.

Normal Distribution

- Data symmetrically distributed
 - no skew
- Symmetric around the mean
- Fully described by:
 - mean (μ)
 - standard deviation (σ)
- All have same shape, but differ in μ and σ

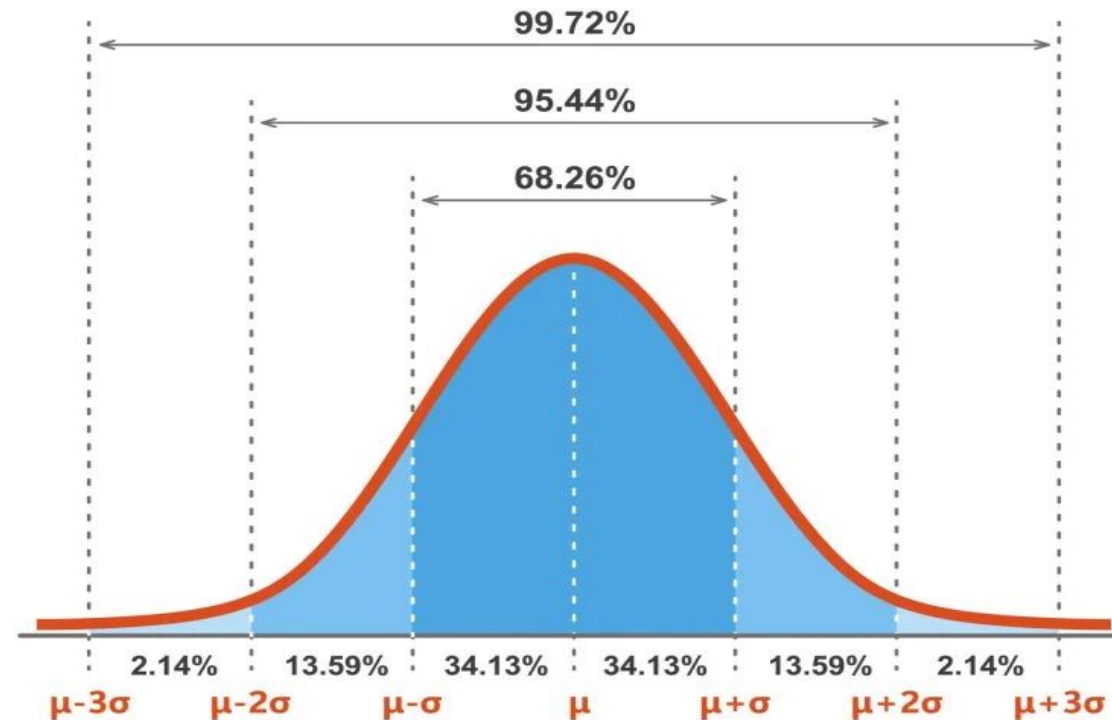


Characteristics of a Normal Distribution

The Empirical Rule (a.k.a. the 68, 95, 99 rule)

- tells you where most of your values lie in a normal distribution

- ~68% of the data falls within **one** standard deviation of the mean
- ~95% falls within **two** standard deviations
- ~99% falls within **three** standard deviations



Normal Distribution Equation

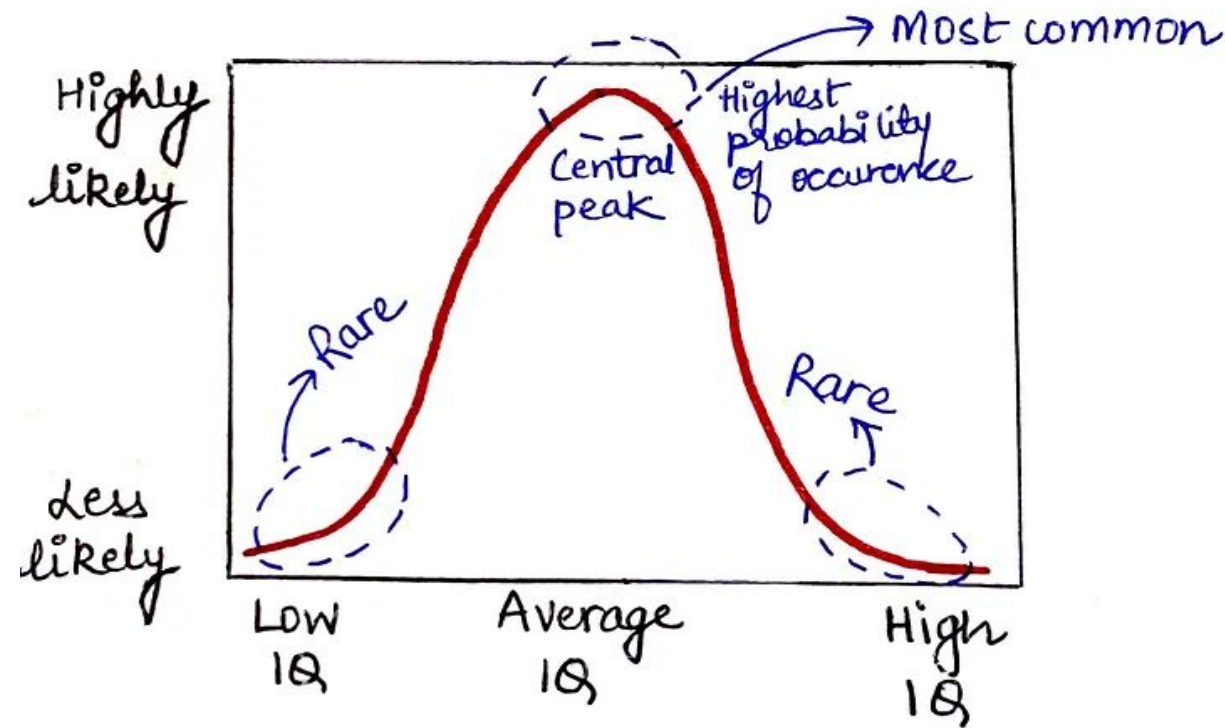
A continuous random variable X is said to have a Normal Distribution if it has a Probability Density Function given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

x	=	observed value
μ	=	mean
σ	=	standard deviation
σ^2	=	variance

Example of a Normal Distribution

IQ levels of a large population follow a normal distribution:

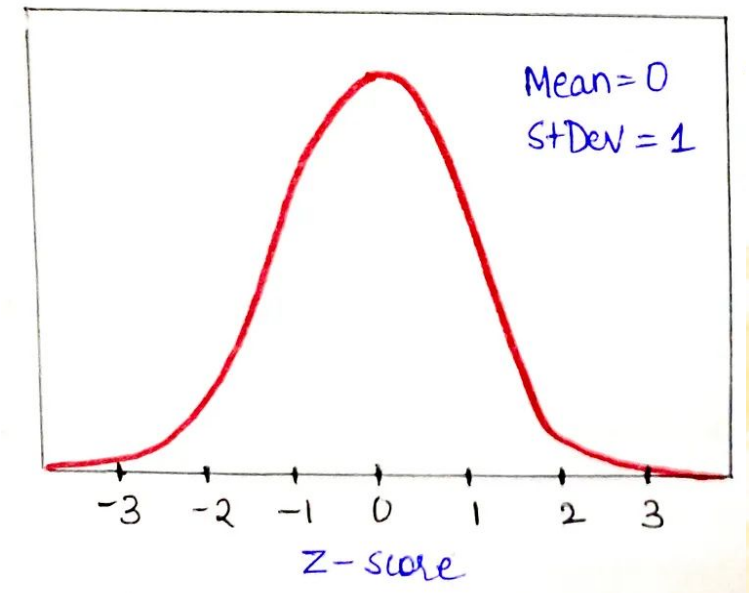


Standard Normal Distribution

- Special case of Normal Distribution, where:
 - $\mu = 0$
 - centered about y axis
 - $\sigma = 1$
 - not too spread out
- Also called the **z-distribution**

Allows for comparison and analysis of data from different distributions by standardizing them into **z-scores**, making it a fundamental tool in statistical analysis.

$$\begin{aligned} f(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \end{aligned}$$



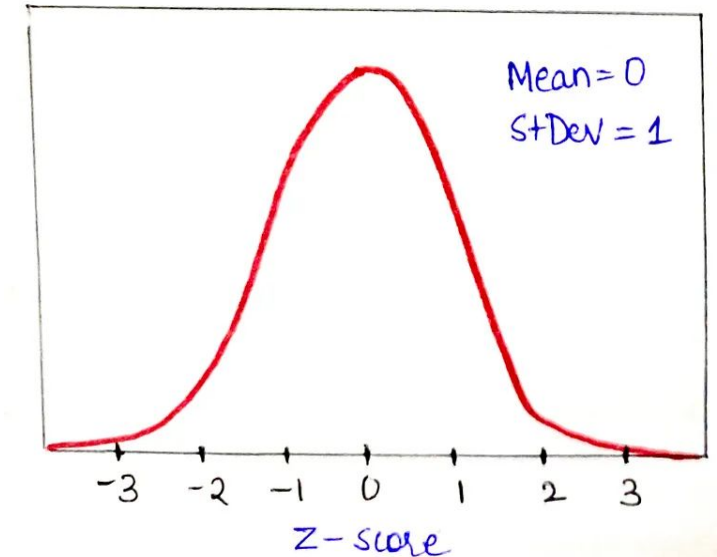
Standard Normal Distribution

- **Any** normal distribution can be **standardized** by converting its values into z scores.
- A **z-score** is a numerical measure of *how many standard deviations a data point is away from the mean* of the distribution.

Calculate Z-Score: subtract the mean and divide by the standard deviation.

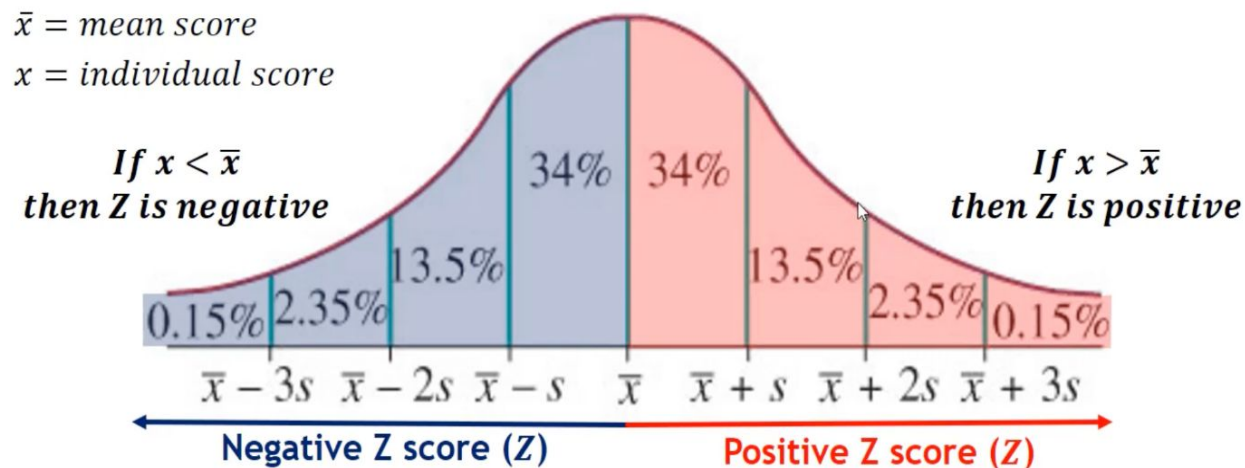
$$Z = \frac{x_i - \mu}{\sigma}$$

Z	=	standard score, or z-score
x_i	=	observed value
μ	=	mean of the sample
σ	=	standard deviation of the sample



Z-Score

- Describes an observed value's **relationship to the mean**
- Measured in terms of **standard deviations** from the mean
- Useful because it allows you to **compare** data points from different distributions by standardizing them



Interpreting Z-Scores:

- **+ z-score:** x value is **greater** than the mean.
- **- z-score:** x value is **less than** the mean.
- **0 z-score:** x value is **equal to** the mean.

Z-Score Example

Calculating Z-Scores: given $\mu = 100$, $\sigma = 15$

If a student scored 115 on a test:

$$Z = (115 - 100) / 15 = 1$$

Interpreting Z-Score: A z-score of +1 indicates that the score is 1 standard deviation above the mean. This indicates that the student performed *better* than the average student, relative to the spread of scores in the distribution.

Benefit: When you standardize a normal distribution, the mean becomes 0 and the standard deviation becomes 1. This allows you to **easily calculate** the probability of certain values occurring in your distribution, or to compare data sets with different means and standard deviations.

Z-Score Example

Example: A math class has a test with a mean score of 80 and a standard deviation of 5 points. The distribution is approximately normally distributed.

Deepak achieves score of 90. What is his Z-Score?

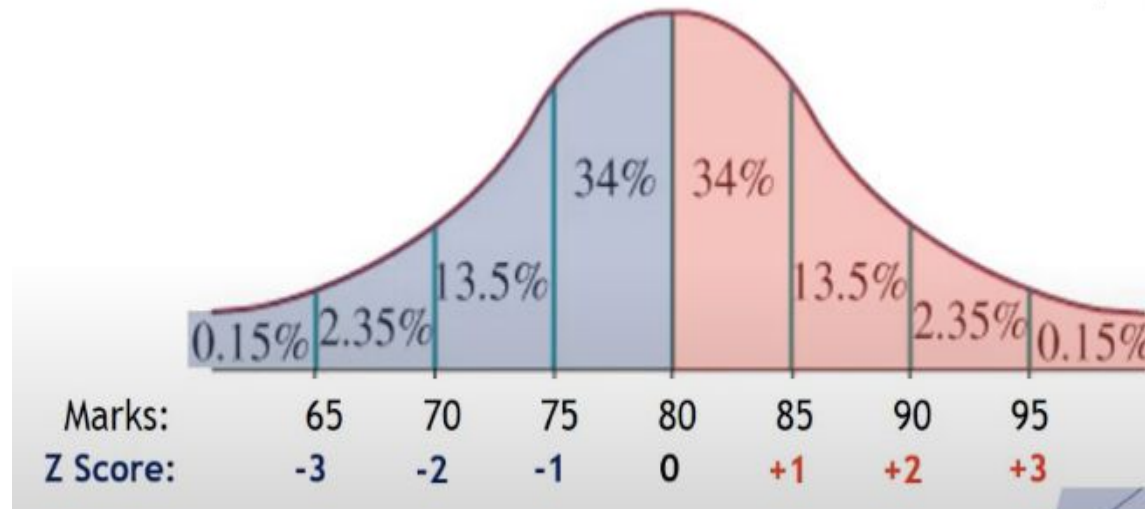
Individual score, $x = 90$

$\mu = 80$

$\sigma = 5$

$Z = ?$

$$Z = \frac{x_i - \mu}{\sigma}$$



Not everything is normal



Other Distributions

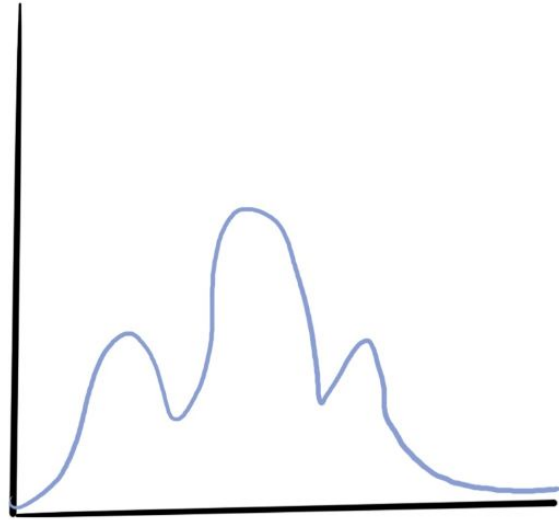
- Binomial Distribution
- Poisson Distribution
- Many more...



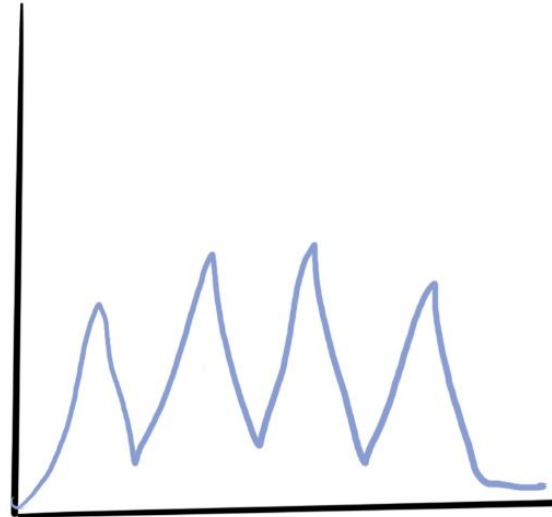
Central Limit Theorem



Comparing two non-normal distributions is hard.



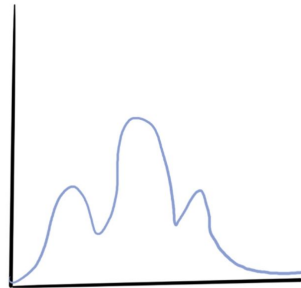
Some annoying distribution



Some other, even more annoying distribution

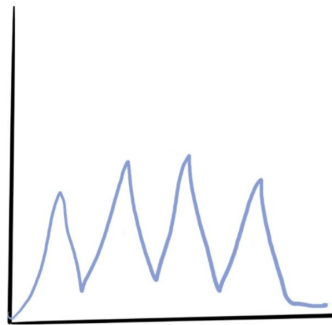
Sampling from non-normal distributions

The two distributions



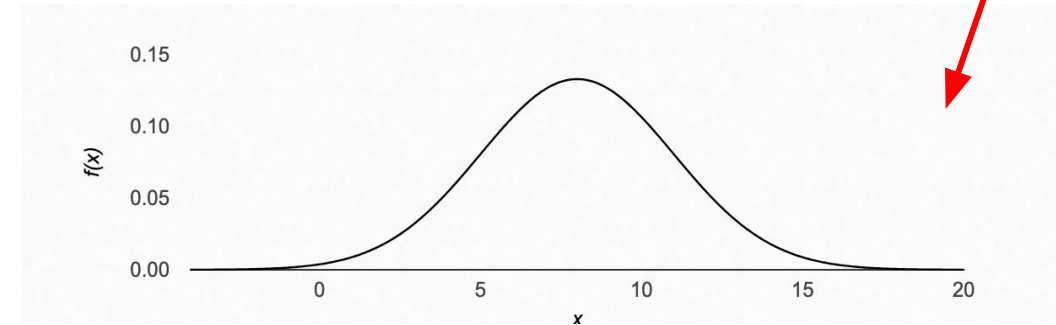
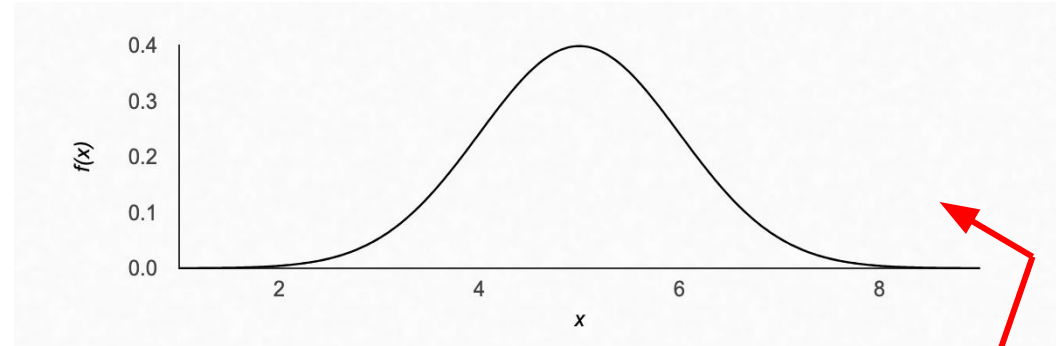
Some annoying distribution

Take a bunch of samples



Some other, even more annoying distribution

Take a bunch of samples



We took a bunch of samples, took the mean of each, and plotted the histograms for each original distribution

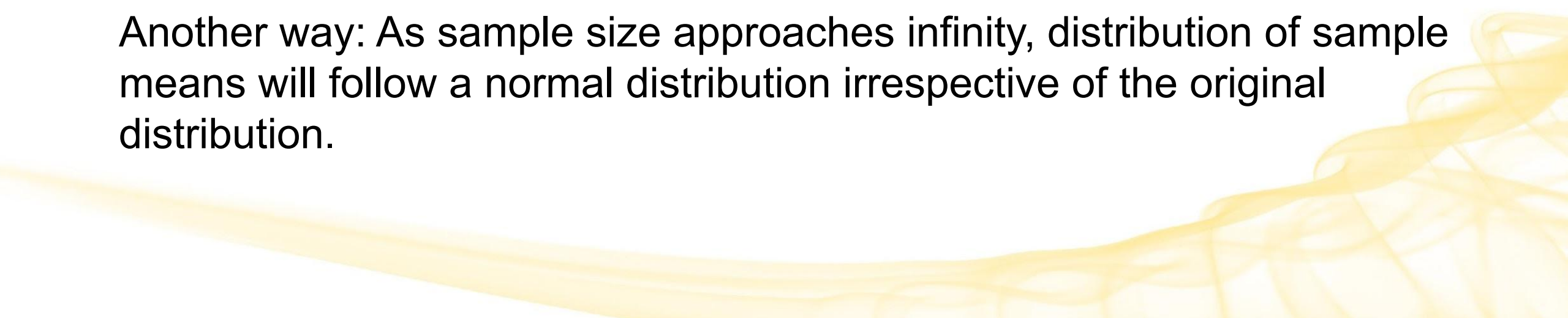
Central Limit Theorem

One of the fundamental theorems in statistics - core to statistical analysis!

States:

“Given a sufficiently large sample size, the sampling distribution of the mean of a variable will approximate a normal distribution regardless of that variable’s distribution in a population.”

Another way: As sample size approaches infinity, distribution of sample means will follow a normal distribution irrespective of the original distribution.



How many is “sufficiently large”?

CLT: “Given a **sufficiently large** sample size, the sampling distribution of the mean of a variable will approximate a normal distribution regardless of that variable’s distribution in a population.”

A sample size is considered "large" if it's at least 30

$n \geq 30 \rightarrow$ “large enough”

However

- Keep in mind that CLT is about a *sampling distribution* but in practice we’d be getting a finite number of samples

And

- If the population is **normal**, then the result holds for samples of any size
 - The sampling distribution of the sample means will be approximately normal even for samples of size less than 30!

Example: Rolling a die



Sampling from a Uniform Distribution

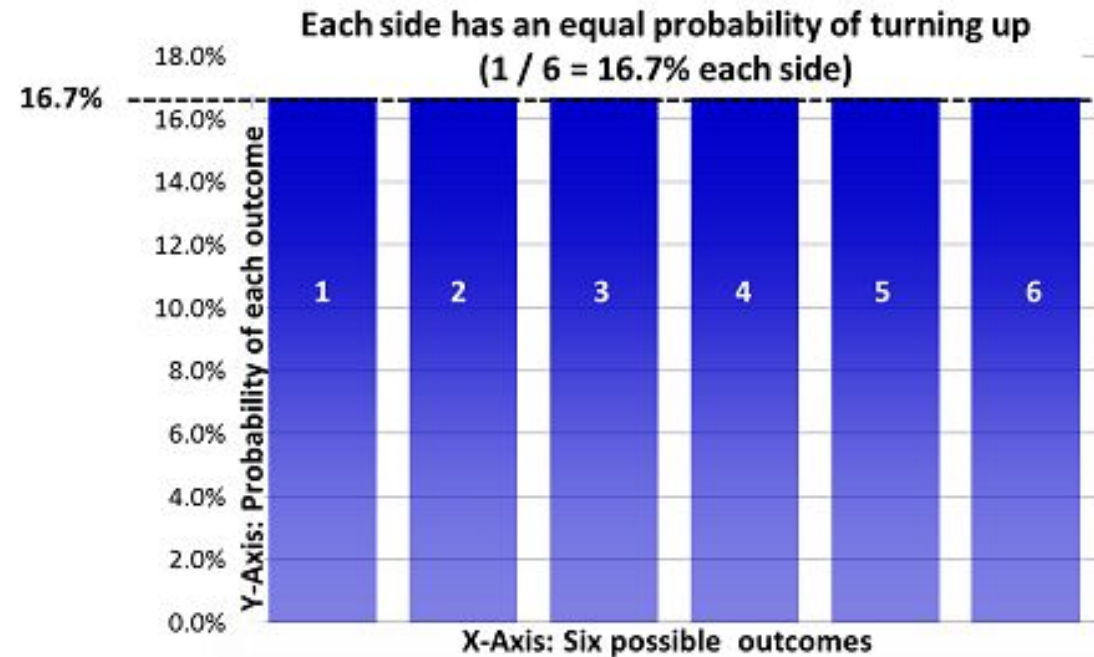
Scenario:

Roll the die 5 times – that's one sample.

Most of the time, the mean will be 3.5

Sometimes, it won't be!

What is the rarest mean?



Example: Back to rolling a die 🎲

If we roll a die several times we might get:

1 2 3 4 5 -> Average of 3

1 1 6 6 3 -> Average of 3.4

1 1 1 1 1 -> Average of 1

6 6 6 6 6 -> Average of 6

We know we have the highest chances of getting 3.5.

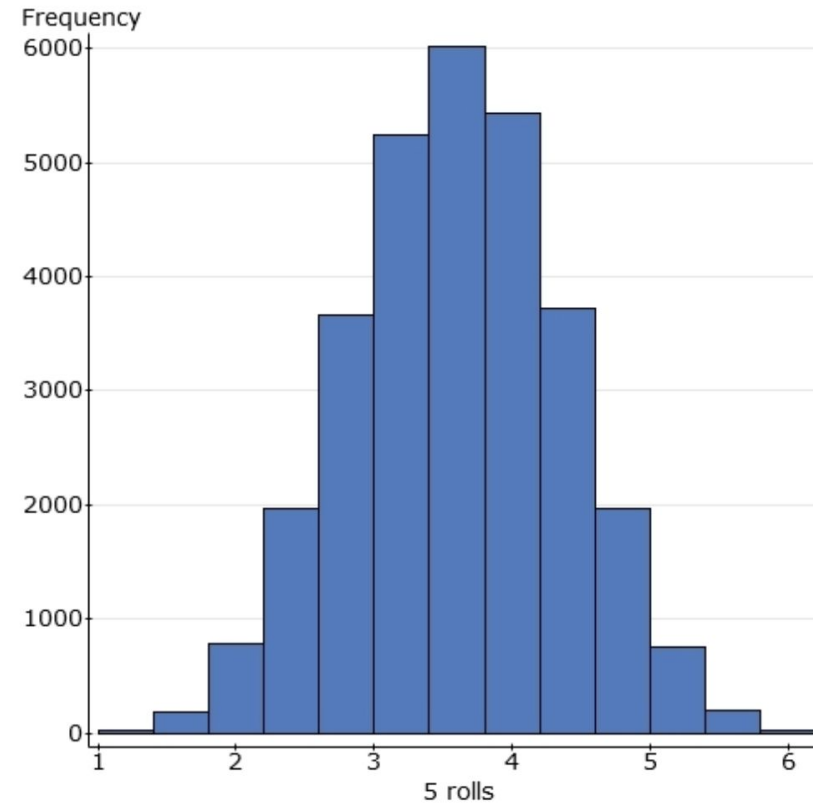
Is there a way we can discuss the likelihood of getting the other means?

Yes. A distribution.



Sampling

If you roll a die five times, here is the frequency with which you will get each of the possible means.

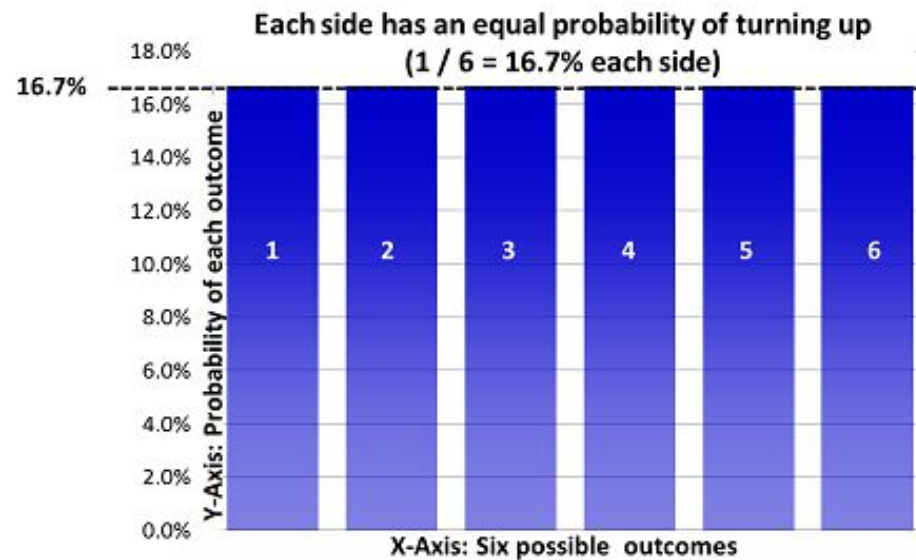


Histogram showing the frequency of each possible mean when rolling a die five times

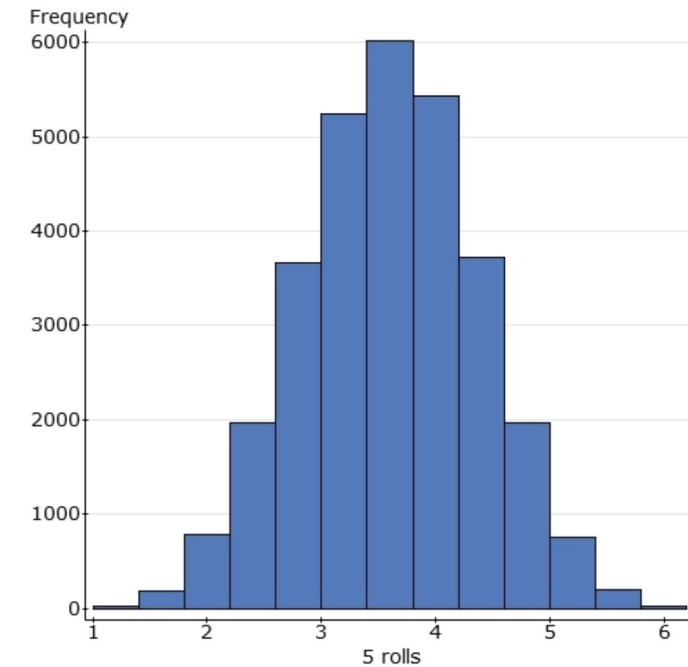
Sampling

According to the Central Limit Theorem, if you roll a die many times and calculate the mean each time, **the sample means will follow a normal distribution.**

We now have two separate things:



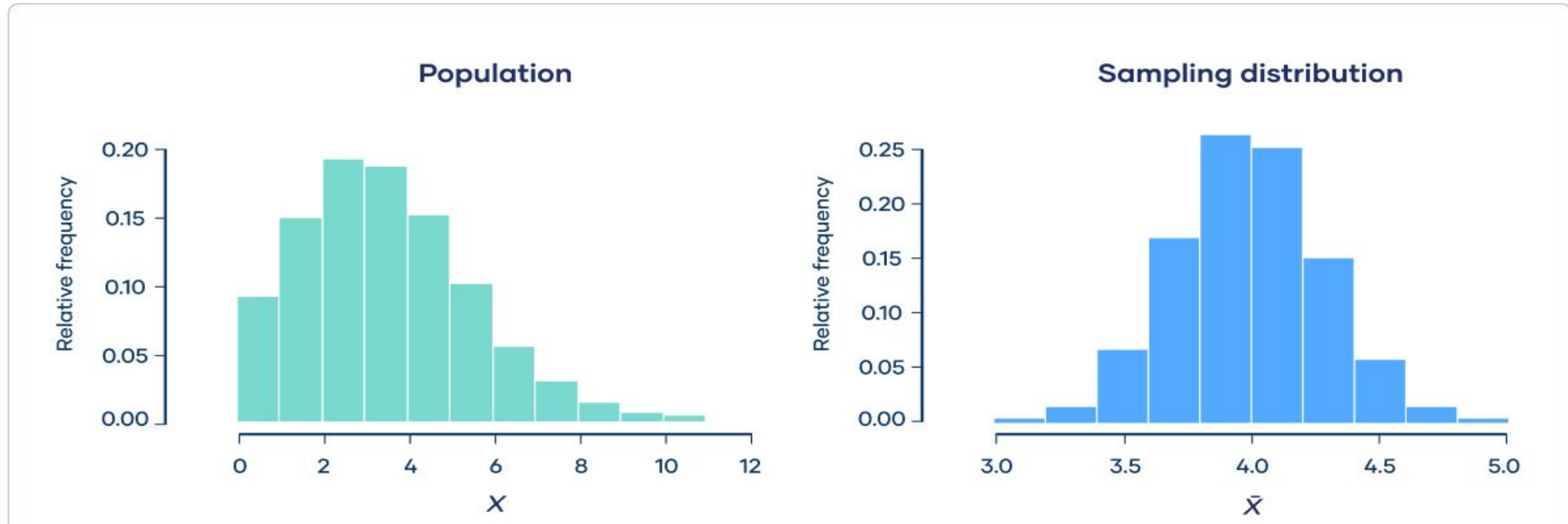
The original distribution



The potential distribution of means when we sample it

Inference: We started with the uniform data distribution, but **the means of samples** drawn from it resulted in a normal distribution.

Another Example: The Central Limit Theorem



A **population** follows a **Poisson distribution** (left image). If we take 10,000 **samples** from the population, each with a sample size of 50, the sample means follow a normal distribution, as predicted by the **central limit theorem** (right image).

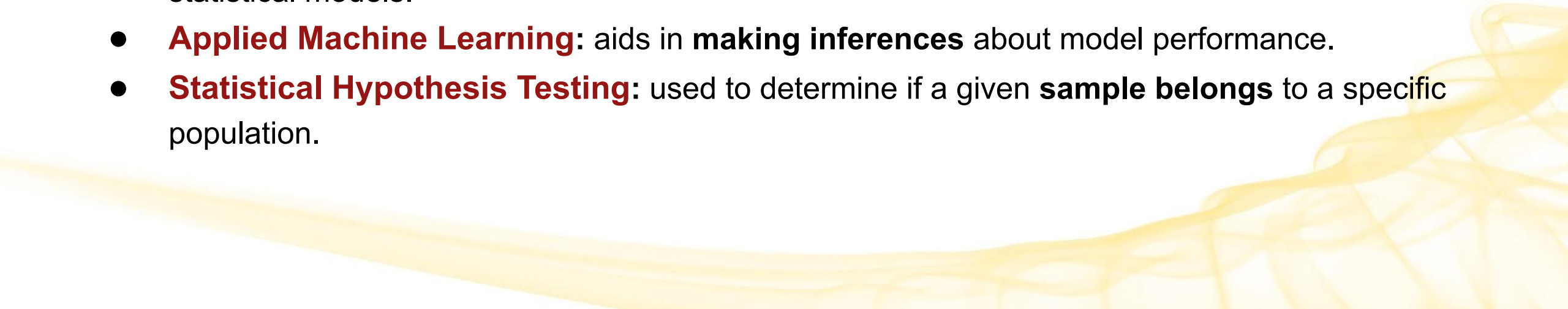


Central Limit Theorem Application

The CLT is useful in various scenarios, particularly when analyzing an entire population is difficult.

- It allows us to assume that the sampling distribution of the mean will be normally-distributed in most cases

Some key applications:

- **Data Science**: helps make **accurate assumptions** about a population to build robust statistical models.
 - **Applied Machine Learning**: aids in **making inferences** about model performance.
 - **Statistical Hypothesis Testing**: used to determine if a given **sample belongs** to a specific population.
- 

Standard Error (SE)

The **Standard Error** of a sample statistic (ex. mean, median) is the **standard deviation** of that statistic's **sampling distribution**, given by:

$$SE = \frac{\sigma}{\sqrt{n}}$$

If the statistic is the sample **mean**, it is called the **Standard Error of the Mean** (SEM).

- SE and SEM are often used interchangeably, though strictly speaking SE could be any sample statistic.
- Mean is most commonly used

If we only have one sample with mean s , we'll estimate the SE like so: $SE \approx \frac{s}{\sqrt{n}}$

Standard Error vs Standard Deviation

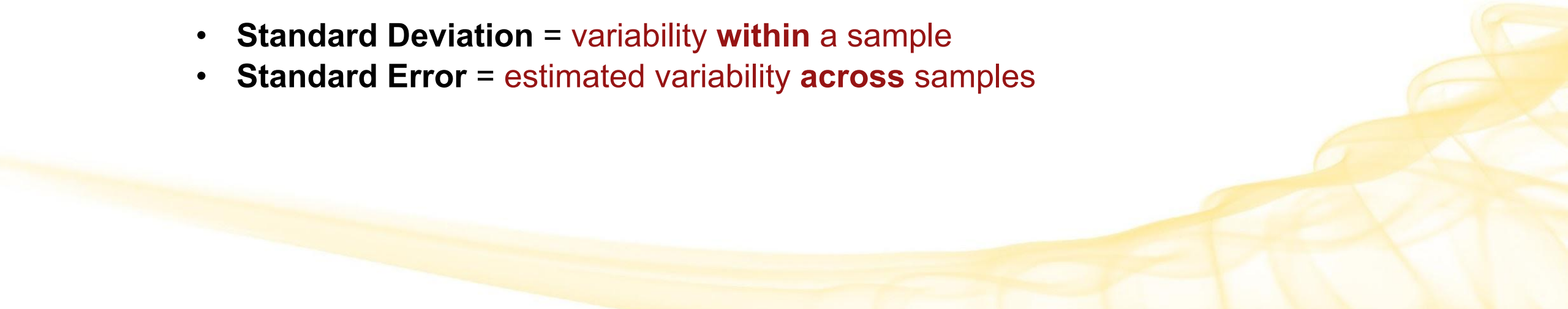
Standard Deviation gives us the variability of values in a sample of the population

ex. how far a data point is likely from the mean

Standard Error gives us the variability of a statistic between samples drawn from the same population

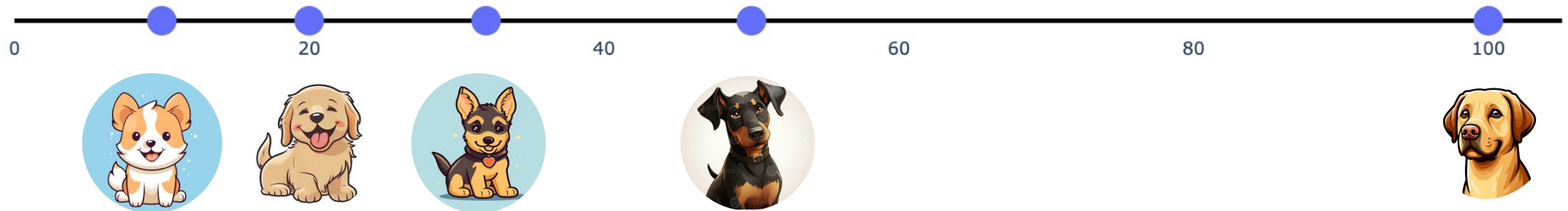
ex. how far a sample mean is likely from the population mean across samples

In other words:

- **Standard Deviation** = variability **within** a sample
 - **Standard Error** = estimated variability **across** samples
- 

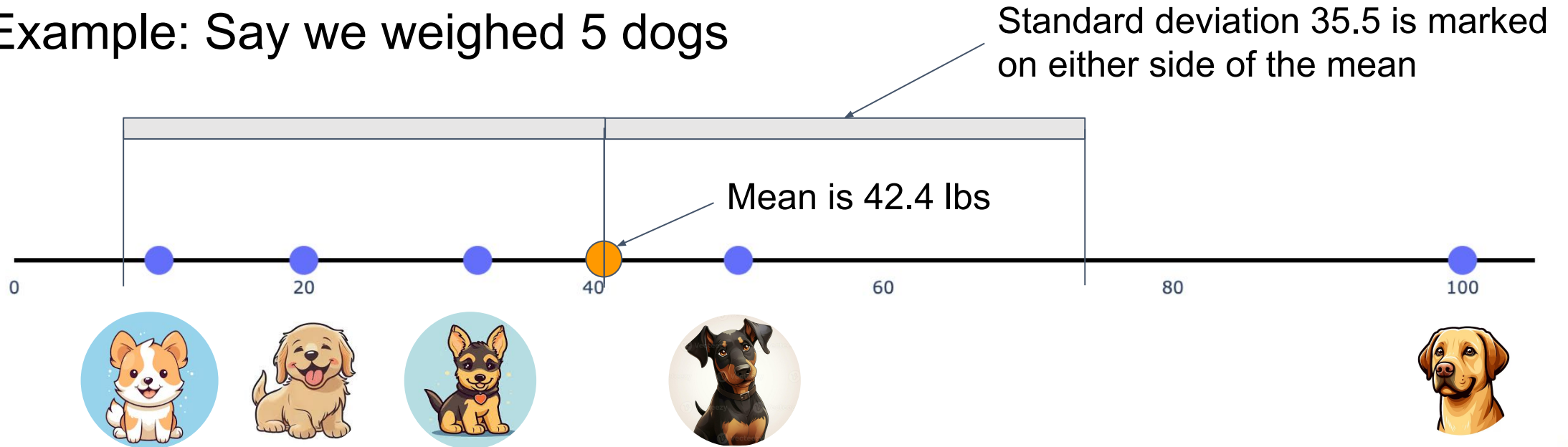
Standard Error vs Standard Deviation

Example: Say we weighed 5 dogs



Standard Error vs Standard Deviation

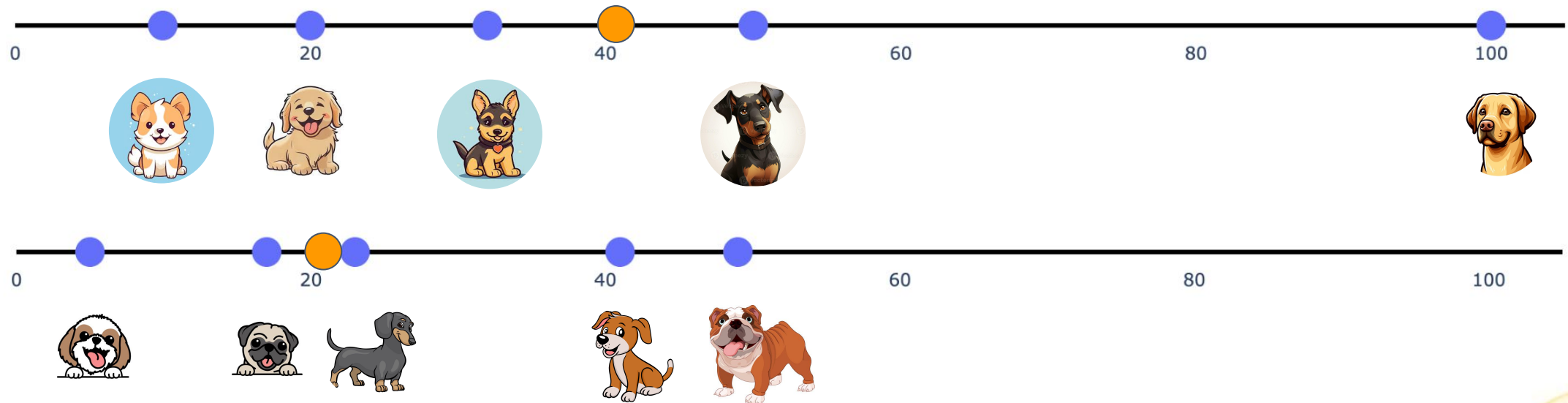
Example: Say we weighed 5 dogs



These measures tell us about how spread out our data is.

Standard Error vs Standard Deviation

Now suppose we did this 5 times with different dogs (25 dogs total)

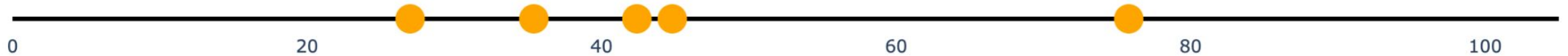


...

We get 5 different sample means, and 5 different sample standard deviations around those means

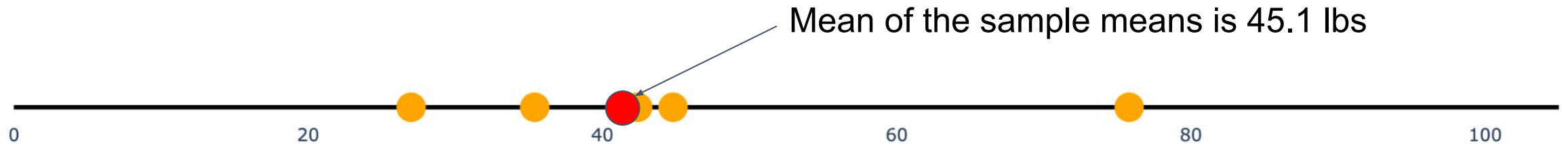
Standard Error vs Standard Deviation

If we plot those 5 sample means on a line, we have



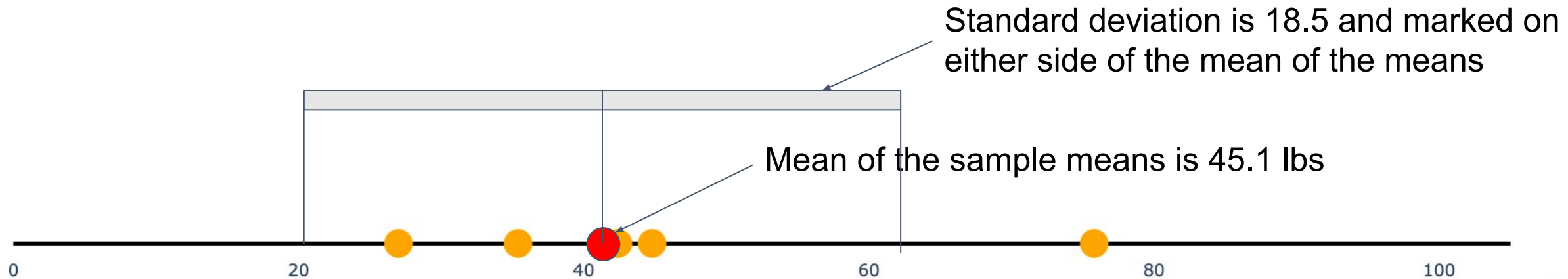
Standard Error vs Standard Deviation

And we can take the mean of those 5 sample means



Standard Error vs Standard Deviation

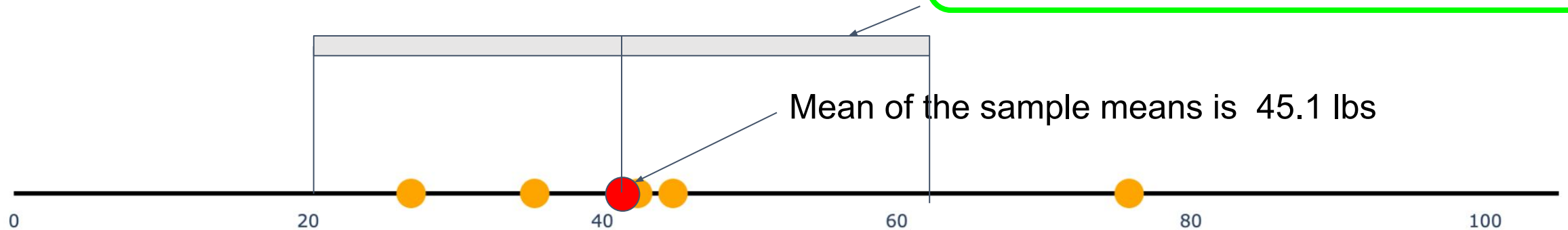
We can calculate the standard deviation of those 5 means



Standard Error vs Standard Deviation

This is the Standard Error

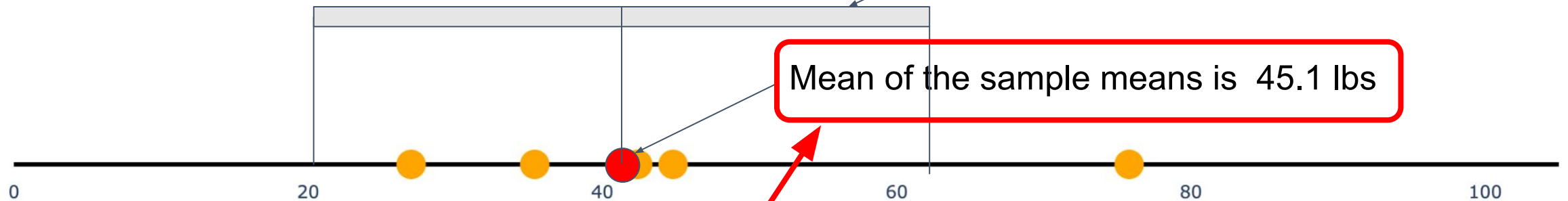
Standard deviation is 18.5 and marked on either side of the mean of the means



Standard Error vs Standard Deviation

If we had infinitely many samples, this would be the Standard Error

Standard deviation is 18.5 and marked on either side of the mean of the means



Mean of the sample means is 45.1 lbs

And recall from CLT that with enough large samples this is a good approximation of the population mean, μ

Standard Error use

You have one sample. The approximated SE tells you how much you would expect the mean to vary if you were to repeat the experiment again, using new samples from the same population.

- Notice that **SE decreases as sample size increases!**
- This makes sense, as bigger samples give us a better approximation of the population

